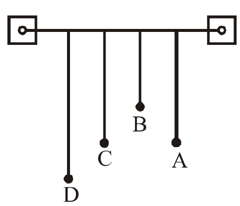
OSCILLATIONS

1. The type of motion of earth about its axis is?
(1) Simple harmonic motion
(2) Periodic but not SHM
(3) Oscillatory
(4) None

2. General vibration of a polyatomic molecule about its equilibrium position is :-
(1) Simple harmonic motion
(2) Neither periodic nor SHM
(3) Superposition of SHMs
(4) None

3. A ball performs uniform circular motion on a horizontal plane then shadow of ball on the vertical wall performs.
(1) Projectile motion
(2) Uniform circular motion
(3) Simple harmonic motion
(4) Non uniform circular motion

4. Four pendulum A.B.C. and D are suspended from the same elastic support as shown in figure. A and C are of same length while B is smaller than A and D is larger than A. If A is given a transverse displacement.



(1) D will vibrate with maximum amplitude
(2) C will vibrate with maximum amplitude
(3) B will vibrate with maximum amplitude
(4) All four will oscillate with equal amplitude

5. A particle oscillating under a force $\vec{F} = -K\vec{x} - b\vec{v}$ is (K and b are constants), then oscillator is :-
(1) Simple harmonic oscillator
(2) Linear oscillator
(3) Damped oscillator
(4) Forced oscillator

6. Time period of oscillation of vertical spring block system is 2 seconds, physical quantity which will be having time period 1sec is :-
(1) Velocity (2) Potential Energy
(3) Acceleration (4) Momentum

7. The equation of motion of a particle is $x = a\cos(\alpha t)^2$ the motion is :-
(1) Periodic but not oscillatory
(2) Periodic and oscillatory
(3) Oscillatory but not periodic
(4) Neither periodic nor oscillatory

8. A body is doing SHM, which is not true :-
(1) Average total energy per cycle is equal to its maximum kinetic energy
(2) Average kinetic energy per cycle in equal to half of its maximum kinetic energy
(3) Mean velocity over a complete cycle in equal to $\frac{2}{\pi}$ times of its max. velocity
(4) Root mean square velocity is $\frac{1}{\sqrt{2}}$ times of its maximum velocity

9. The displacement of a particle varies with time according to the relation $y = a \sin\omega t + b\cos\omega t$
(1) The motion is oscillatory but not SHM
(2) The motion is SHM with amplitude a + b
(3) The motion is SHM with amplitude a^2+b^2
(4) The motion is SHM with amplitude $\sqrt{a^2+b^2}$

10. In SHM when the magnitude of displacement is greatest, then :-
(1) Magnitude of velocity is least
(2) Magnitude of velocity is greatest
(3) Magnitude of acceleration is least
(4) Magnitude of force is least

11. In SHM phase difference between displacement and velocity is :-
(1) Zero (2) π
(3) $\pi/2$ (4) None of these

12. In damped oscillation mechanical energy decreases with time :-
(1) Linearly (2) Exponentially
(3) Not decreases (4) None of these

13. In the ideal case of zero damping the amplitude of SHM at resonance is :-

(1) Minimum

(2) Zero

(3) Infinite

(4) Maximum but not infinite

14. A displacement equation of a particle executes S.H.M is $x = \sin^2\omega t$, then :-

(a) Mean position $x = 0$

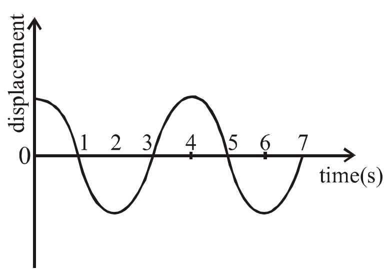
(b) Mean position $x = \frac{1}{2}$

(c) Time period $\frac{\pi}{\omega}$

(d) Time period $= \frac{2\pi}{\omega}$

(1) a, d (2) b,c (3) a,c (4) b,d

15. Displacement vs time curve for a particle executing S.H.M. is shown in fig given below. Choose the correct statement :-



(1) Phase of the oscillation is same at t = 0s and t = 2s

(2) Phase of the oscillation is same at t = 2s and t = 6s

(3) Phase of the oscillation is same at t = 1s and t = 7s

(4) Phase of the oscillation is same at t = 1s and t = 3 s

6. Which one of the following is not a periodic motion ?

(1) Rotation of the earth about its axis

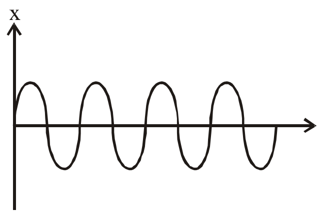
(2) A freely suspended bar magnet displaced from its N-S direction and released

(3) Motion of hands of a clock

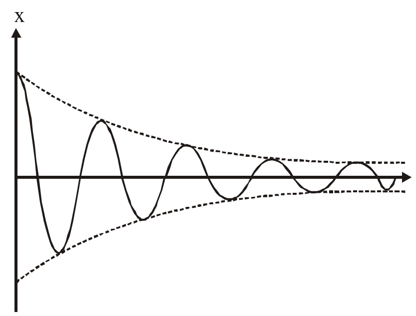
(4) An arrow released from a bow

17. Which of the following displacement time graphs represent under damped harmonic oscillations?

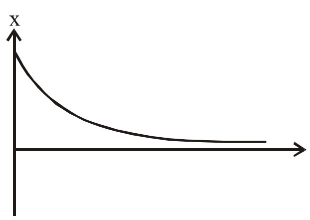
(1)



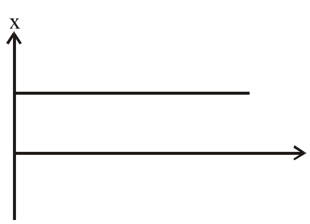
(2)



(3)



(4)



18. In case of forced oscillations of a body :-

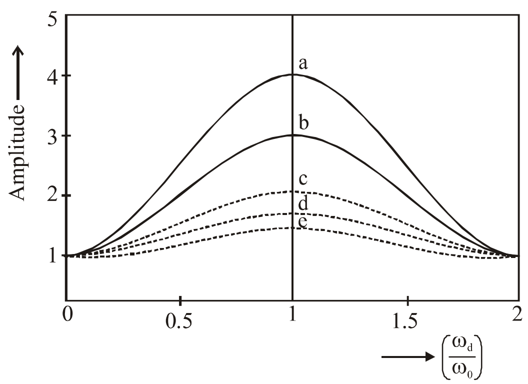
(1) Driving force is constant

(2) Driving force is to be applied only moment arily

(3) Driving force has to be periodic and continuous

(4) Driving force is not required

19. Graph below shows variation of amplitude of forced oscillations with respect to frequency of driving force. Choose the correct option.



(1) Amplitude is maximum at $\omega_d = \omega_0$

(2) Peak amplitude is maximum for curve (a) because for curve (a) damping is minimum

(3) In case of resonance there occur maximum transfer of energy from driver to driven

(4) All of above

20. The natural frequency of building depend on :-

(a) Its height and other size parameter

(b) The nature of building material

The correct statement :-

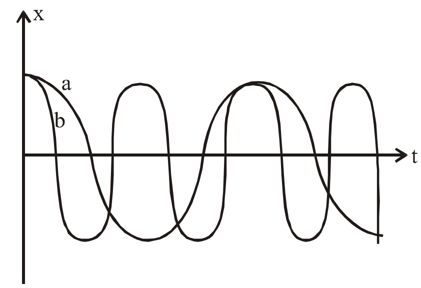
(1) Only (a)

(2) Only (b)

(3) Both (a) & (b)

(4) Neither (a) & (b)

21. Choose incorrect option :-



(1) Amplitude of both are same

(2) Both curve have same frequency

(3) Initial phase of both curve are same

(4) Time period of a is twice of time period of b

22. In SHM :-

(1) PE is stored due to elasticity of system

(2) KE is stored due to inertia of system

(3) Total energy of system remains conserved

(4) All of above

23. A particle is acted simultaneously by mutually perpendicular simple harmonic motion $x = a\cos\omega t$ and $y = a\sin\omega t$. The trajectory of motion of the particle will be :-

(1) an ellipse (2) a parabola

(3) a circle (4) a straight line

24. Which of the following is/are not true for a simple harmonic oscillator ?

(1) Force acting is directly proportional to displacement from the mean position and opposite to it

(2) Motion is periodic

(3) Acceleration of the oscillator is constant

(4) Velocity is periodic